# Xian-Zhang Chen

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## Education

- **ICPCS, Lanzhou University** — Lanzhou, China
  *Ph.D. in Theoretical Physics* — *September 2016 - June 2023*

  *Thesis title: Wavefunction-related Quantum Transport and Many-body effects in Graphene*
  *Advisor: Prof. Liang Huang and Prof. Rodolfo A. Jalabert*

- **IPCMS, University of Strasbourg** — Strasbourg, France
  *Joint training Ph.D. student* — *December 2020 - November 2022*

  *Project title: Theory of Scanning Gate Microscopy in Graphene*
  *Advisor: Prof. Rodolfo A. Jalabert*

- **School of Physical Science and Technology, Lanzhou University** — Lanzhou, China
  *Bachelor of Theoretical Physics* — *2012 - 2016*

  *Thesis title: Quantum Transport in an Octagonal Graphene dot*
  *Advisor: Prof. Liang Huang*

## Employment

- **Ningbo Institute of Digital Twin / EIT** — Ningbo, China
  *Research Assistant Professor* — *December 2025 - Present*

  *Research field: Topological Quantum Computing and Novel Magnetism*
  *Cooperative Advisor: Prof. Tong Zhou*

- **EIAS, Eastern Institute of Technology, Ningbo** — Ningbo, China
  *PostDoc in Condensed Matter Theory* — *September 2023 - November 2025*

  *Project title: Topological Quantum Computing in Magnetic textures*
  *Advisor: Prof. Tong Zhou*

## Honors and Awards

- Selected as a member of Programme Doctoral International (PDI) of Collège doctoral Européen (CDE), 2021
- China Scholarship Council award (EUR 32,400), China Scholarship Council, 2020
- Excellent Poster Award, The 5th National Statistical Physics Conference, 2019
- National Scholarship for Postgraduates (CNY 20,000), Ministry of Education of P. R. China, 2016

## Personal Statement

My research centers on magnetic textures and topological quantum devices. I develop skyrmion-engineered platforms to generate and control Majorana zero modes, and I design braiding and fusion protocols driven by skyrmion motion to realize non-Abelian statistics. To address readout, I propose scanning-gate-microscopy (SGM)–based detection approaches that interface naturally with transport. In a complementary line, I investigate altermagnetism, mapping its topological characteristics, analyzing proximity effects, and linking altermagnetic phases to topological superconductivity. My simulations combine KWANT (quantum transport/BdG), OOMMF (skyrmion micromagnetics), and DFT (altermagnet electronic structures).
Prior to this, my Ph.D. at ICPCS, Lanzhou University examined wavefunction properties in graphene—including SGM, magnetotransport, and many-body spectral statistics—with an emphasis on experimentally relevant theory and numerics for Dirac quasiparticles. This training established my habit of building end-to-end, experiment-proximate models, which I now apply to skyrmion-based topological devices and altermagnetic materials.

**Research Interests**:

- Topological Quantum Computing
- Skyrmions
- Altermagnet
- Scanning Gate Microscopy Technology
- Many-body Physics

## Publications and Preprints

### Peer-reviewed publications

**[9]** Ziye Zhu, Richang Huang, **X.-Z. Chen**, Xunkai Duan, Jiayong Zhang, Igor Zutic, Tong Zhou, *Altermagnetic Proximity Effect*, Phys. Rev. Lett. 136, 186702 (2026).

*Contribution: Model discussion and manuscript review.*

**[8]** Zhou Cui, Ziye Zhu, Xunkai Duan, Bowen Hao, **X.-Z. Chen**, Jiayong Zhang, Tong Zhou, *Symmetry-Driven Unconventional Magnetoelectric Coupling in Perovskite Altermagnets: From Bulk to the Two-Dimensional Limit*, Advanced Science, e00004 (2026).

**[7]** Sihui Wu, Jiahui Qian, Xunkai Duan, **X.-Z. Chen**, Zhou Cui, Jiayong Zhang, Chunlan Ma, Tong Zhou, *Electrically switchable quantum anomalous and spin Hall states in two-dimensional honeycomb lattices with non-Dirac bands*, Phys. Rev. B 111, 235414 (2025).

*Contribution: Model discussion and manuscript review.*

**[6]** **X.-Z. Chen**, Guillaume Weick, Dietmar Weinmann, Rodolfo A. Jalabert, *Scanning gate microscopy in graphene nanostructures*, Phys. Rev. B 107, 085420 (2023).

*Contribution: Analytical derivation, numerical calculations and analysis, figures, and manuscript development.*

**[5]** **X.-Z. Chen**, Zhen-Qi Chen, Liang Huang, Celso Grebogi, Ying-Cheng Lai, *Many-body spectral statistics of relativistic quantum billiard systems*, Phys. Rev. Research 5, 013050 (2023).

*Contribution: Initial manuscript writing, numerical calculations and analysis, and figures.*

**[4]** Xiao-Liang Li, **X.-Z. Chen**, Chen-Rong Liu, Liang Huang, *Quantization condition of scarring states in complex soft-wall quantum billiards*, Acta Phys. Sin. 69, 080506 (2020).

*Contribution: Assistance with numerical programs, analysis, and figures.*

**[3]** Guan-Qun Zhang, **X.-Z. Chen**, Li Lin, Hai-Lin Peng, Zhong-Fan Liu, Liang Huang, Ning Kang, Hong-Qi Xu, *Transport signatures of relativistic quantum scars in a graphene cavity*, Phys. Rev. B 101, 085404 (2020).

*Contribution: Co-first author; numerical simulations, analysis of experimental results, and supplementary material writing.*

**[2]** Chen-Rong Liu, Pei Yu, **X.-Z. Chen**, Hong-Ya Xu, Liang Huang, Ying-Cheng Lai, *Enhancing von Neumann entropy by chaos in spin-orbit entanglement*, Chin. Phys. B 28, 100501 (2019).

*Contribution: Numerical-results discussion and manuscript review.*

**[1]** Chen-Rong Liu, **X.-Z. Chen**, Hong-Ya Xu, Liang Huang, Ying-Cheng Lai, *Effect of chaos on two-dimensional spin transport*, Phys. Rev. B 98, 115305 (2018).

*Contribution: Assistance with numerical programs and discussion of numerical results.*

### Preprints

**[P3]** **X.-Z. Chen**, Jiayong Zhang, Bowen Hao, Jiahui Qian, Ziye Zhu, Igor Zutic, Zhenyu Zhang, Tong Zhou, *Altermagnets Enable Gate-Switchable Helical and Chiral Topological Transport with Spin-Valley-Momentum-Locked Dual Protection*, arXiv:2603.06487.

**[P2]** Ziye Zhu, **X.-Z. Chen**, Xunkai Duan, Zhou Cui, Jiayong Zhang, Igor Zutic, Tong Zhou, *Altermagnetoelectric Spin Field Effect Transistor*, arXiv:2512.02974.

*Contribution: Transport calculations and partial manuscript writing.*

**[P1]** Richang Huang, Yongliang Hu, **X.-Z. Chen**, Peng Yu, Igor Zutic, Tong Zhou, *Towards Scalable Braiding: Topological Superconductivity Unlocked under Nearly Arbitrary Magnetic Field Directions in Planar Josephson Junctions*, arXiv:2504.20031.

*Contribution: Assistance with figures and manuscript discussion and review.*

## Presentations

- **Talks**
  - Novel quantum spin and quantum anomalous Hall effects based on altermagnet
    CPS Meeting 2025, Harbin, China, 09/2025.
  - Many-body spectral statistics of relativistic quantum billiards systems
    DPG Meeting 2022, Regensburg, Germany, 09/2022.
- **Posters**
  - Transport signatures of scars in graphene dots at low magnetic field
    GDR 2426, Aussois, France, 12/2021.
  - Many-body spectral statistics of relativistic quantum billiard systems
    The 5th National Statistical Physics Conference (SPCSC), Hefei, China, 07/2019.
  - Transport signatures of relativistic quantum scars in a graphene cavity
    International Workshop on Experimental and Theoretical Developments on Complex Quantum Systems, Lanzhou, China, 07/2018.

## Participated Grants

- Spectral statistics of a class of pseudo-integrable quantum billiards, CNY 630,000, No. 12175090, NSFC, 2021.
- Study of spin anomalous effects in Dirac billiards, CNY 580,000, No. 11775101, NSFC, 2017.